Audit Trail — Case 11: Shoulder closure on divided highway

Field	Value
Plan status	2 changes · 1 needs attention · 13 checked · 2 pending · reference
MUTCD typical application	TA-3
CDOT standard sheet	S-630-1
Case	Case 11: Shoulder closure on divided highway
Taper length	217 ft (L/3 (shoulder taper))
Buffer space	645 ft
Device spacing — taper	65 ft
Device spacing — tangent	130 ft
Crew steps	11
Routing	shoulder_no_reduction

Taper Length

Speed 65 mph >= 45 mph threshold -> using L = W x S

L = 10 x 65 = 650 ft

L/3 = 650 / 3 = 217 ft

Field	Value
Closure type	shoulder
Offset (shoulder width)	10 ft
Required taper (L/3 (shoulder taper))	217 ft

Source: MUTCD 11th Ed. Sec 6B.08, Table 6B-4 (taper length L). Shoulder closures use L/3 per Sec 6B.08 (Table 6B-3).

CDOT reference: CDOT S-630-1 Case 11 (right-shoulder closure on divided highway)

Buffer Space

MUTCD Table 6B-2: 645 ft. CDOT S-630-1 Sheet 14 posts a 570 ft minimum at 65 mph, but only as part of the Case 26 mandatory speed step-down (65 -> 60 mph); this plan has no qualifying reduction, so the federal posted-speed value governs (Sheet 7 Case 11: buffer VARIES per General Note 24).

Source: MUTCD 11th Ed. Sec 6B.06, Table 6B-2 (stopping sight distance). CDOT S-630-1 Sheet 7 Case 11 (buffer VARIES per General Note 24).

Channelizing Device Spacing

65 mph -> 65 ft spacing (MUTCD Sec 6K.01: spacing equals speed in feet)

65 mph -> 130 ft spacing (MUTCD Sec 6K.01: spacing equals 2x speed in feet)

L/3 = 217 ft / 65 ft max spacing -> 5 drums at 54.2 ft intervals

1000 ft / 130 ft max spacing -> 9 cones at 125.0 ft intervals

Downstream taper: 50 ft at ~25 ft spacing -> 2 cones (MUTCD Sec 6B.08: 50-100 ft downstream taper); 9 tangent + 2 downstream = 11 cones total

Source: MUTCD 11th Ed. Sec 6K.01

Advance Warning Signs

Speed 65 mph, road_type = 'rural' -> Table 6B-1 category: rural

A = 500 ft, B = 500 ft, C = 500 ft

Position	Code	Station (ft)	Distance from Taper (ft)
C (furthest)	W20-1	3,362	1,500 upstream
B (middle)	W20-2	2,862	1,000 upstream
A (nearest)	W21-5aR	2,362	500 upstream

Source: MUTCD 11th Ed. Table 6B-1

Colorado Requirements (CDOT S-630-1)

Check	Result	Citation	Detail
Signs on both sides of divided highway	FAIL	CDOT S-630-1 (July 2026) Sheet 2, General Note 8	Required: True. Signs placed: 7 left, 11 right.
G20-5P Work Zone signs every 2,640 ft	PASS	CDOT S-630-1 (July 2026) Sheet 2, General Note 4	Zone length: 3,362 ft. Required: 2. Placed: 2.
Speed reduction <= 15 mph per sign installation	PASS	CDOT S-630-1 (July 2026) Sheet 2, General Note 3	No work-zone speed reduction. Posted speed 65 mph applies throughout the zone.
Flagger station lighting 500W @ 8 ft	PASS	CDOT S-630-1 (July 2026) Sheet 2, General Note 22	Not applicable (no flaggers).

AADT threshold for mobile operations (<= 2,000) — Not applicable (not a mobile operation). (CDOT S-630-1 (July 2026) Sheet 23, Case 38 Note 1)

All Colorado checks pass: False

CDOT Case Reference

Case 11: Shoulder closure on divided highway

This scenario matches CDOT Standard Plan S-630-1, Case 11: shoulder closure on a divided highway.

The generated plan follows the same device layout as the reference case with spacing computed for 65 mph.

Routing: shoulder_no_reduction

Reference:
https://www.codot.gov/safety/traffic-safety/assets/s-standard-plans/s-630-1/S-630-01%20(19-Page%20Set).pdf

Geometry Validation

Work zone 1000 ft vs required taper 217 ft (L/3 (shoulder taper)) + buffer 645 ft.

All geometry checks pass: True

Source: MUTCD 11th Ed. Sec 6B.06 (buffer) and Sec 6B.08 (taper)

Corridor Validation

OpenStreetMap corridor check ran with no warnings.

Site Conditions

Scanned along the corridor against OpenStreetMap at 2026-09-08T16:18:40+00:00 (7147 ms, memoised, via https://overpass-api.de/api/interpreter). The five rule-bearing conditions below are plan facts; a detected condition fired the matching Site Adjustment that follows.

Condition	Result	Evidence
Adjacent at-grade intersection	DETECTED	46 found · nearest 95.8 ft from anchor · unnamed at 39.7256, -104.9875 [work_zone @ 460 ft]
Adjacent interchange (highway ramps)	DETECTED	10 found · nearest 10.8 ft from anchor · North Broadway [downstream @ -3 ft]
Pedestrian sidewalks	DETECTED	103 found · nearest 100.1 ft from anchor · unnamed at 39.7269, -104.9869 [lateral 123 ft off centerline]
Bike lane / cycleway	DETECTED	11 found · nearest 109.6 ft from anchor · unnamed at 39.7267, -104.9868 [lateral 133 ft off centerline]
School zone	None along the corridor	
Railroad crossing	Reference — none	
Hospital	Reference — none	
Road curvature	Reference — none	

Site Adjustments

Site-condition flags layered onto the baseline MUTCD/CDOT layout. Each adjustment is traced to its source rule; the rendered plan, device list, and crew narrative reflect every item below.

Rule	Adjustment
MUTCD §6N.12 p. 848 — Work within the Traveled Way at an Intersection (11th Ed.)	No devices added — the cross-street approach layout is not generated; see the pending-verification disclosure.
MUTCD §6N.16 p. 851 — Interchanges (11th Ed.)	No devices added — the per-ramp interchange layout is not generated; see the pending-verification disclosure.
MUTCD §6C.02 — pedestrian considerations in work zones	Added 4 Type III barricades (sidewalk closure points) and 2 R9-9 SIDEWALK CLOSED signs at the upstream and downstream ends of the work zone.
MUTCD §9C.101 — bicycle facility work zone signing	Added 2 M4-9a BIKE DETOUR signs at the upstream and downstream ends.

Flagger Control

Not applicable for this scenario.

Source: MUTCD 11th Ed. Chapter 6D (Flagger Control) and Chapter 6E (One-Lane, Two-Way Traffic Control).

Pending Verification

2 reference(s) pending verification. An adjacent intersection is flagged, but Conestruct does not generate the cross-street approach layout or evaluate traffic-signal operation, and adds no devices for it. Cross-street control design and the signal-operation review (MUTCD §6N.12, 11th Ed. p. 848; Part 4 for signal-head visibility) remain with the traffic control supervisor. Intersection support is tracked at issue #117; corner-quadrant work at issue #128. Tracking: https://github.com/rtmakatura/conestruct/issues/117

- [intersection_layout_not_generated] An adjacent intersection is flagged, but Conestruct does not generate the cross-street approach layout or evaluate traffic-signal operation, and adds no devices for it. Cross-street control design and the signal-operation review (MUTCD §6N.12, 11th Ed. p. 848; Part 4 for signal-head visibility) remain with the traffic control supervisor. Intersection support is tracked at issue #117; corner-quadrant work at issue #128. (https://github.com/rtmakatura/conestruct/issues/117)
- [interchange_layout_not_generated] An adjacent interchange is flagged, but Conestruct does not generate the per-ramp layout and adds no devices for it. Ramp-specific control design and the ramp-access and coordination review (MUTCD §6N.16, 11th Ed. p. 851) remain with the traffic control supervisor. Tracked at issue #117. (https://github.com/rtmakatura/conestruct/issues/117)